TOR MODULES OF AN AFFINE BLOWUP ALGEBRA ARE ANNIHILATED BY A POWER OF THE IDEAL IN DEGREES BEYOND THE NUMBER OF GENERATORS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-local-cohomology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The affine blowup algebra of an ideal with respect to one of its elements, described as a quotient by the affine blowup relations modulo torsion. Invoked inside the counted proof at source lines 4562-4566 and attributed to a lemma of the cited chapter (a cross-chapter pointer). Provides the presentation of the algebra whose Tor modules the statement bounds.
- The Koszul complex of the blowup relations and the identification of Tor with the homology of the derived tensor product. Used inside the counted proof at source lines 4566-4602. Reduces the annihilation statement to a statement about the homology of an explicit complex in high degrees.

Source reference. Original reference algebra-lemma-affine-blowup-quotient-description: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 4566–4566. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-affine-blowup-quotient-description, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference derived-lemma-trick-vanishing-composition: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 3675–3675. Navigation pointer only: the counted text refers to the Stacks item derived-lemma-trick-vanishing-composition, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:local-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $I = (a_1, \ldots, a_t)$ be an ideal of a Noetherian ring A. Set $a = a_1$ and denote $B = A[\frac{I}{a}]$ the affine blowup algebra. There exists a $c > 0$ such that $Tor_i^A(B, M)$ is annihilated by I^c for all A-modules M and $i \geq t$.*

Proof. Recall that B is the quotient of $A[x_2, \ldots, x_t]/(a_1x_2 - a_2, \ldots, a_1x_t - a_t)$ by its a_1-torsion, see Algebra, Lemma algebra-lemma-affine-blowup-quotient-description. Let

$$B_\bullet = \text{Koszul complex on } a_1x_2 - a_2, \ldots, a_1x_t - a_t \text{ over } A[x_2, \ldots, x_t]$$

viewed as a chain complex sitting in degrees $(t-1), \ldots, 0$. The complex $B_\bullet[1/a_1]$ is isomorphic to the Koszul complex on $x_2 - a_2/a_1, \ldots, x_t - a_t/a_1$ which is a regular sequence in $A[1/a_1][x_2, \ldots, x_t]$. Since regular sequences are Koszul regular, we conclude that the augmentation

$$\epsilon : B_\bullet \longrightarrow B$$

is a quasi-isomorphism after inverting a_1. Since the homology modules of the cone $C_\bullet$ on ϵ are finite $A[x_2, \ldots, x_n]$-modules and since $C_\bullet$ is bounded, we conclude that there exists a $c \geq 0$ such that a_1^c annihilates all of these. By Derived Categories, Lemma derived-lemma-trick-vanishing-composition this implies that, after possibly replacing c by a larger integer, that a_1^c is zero on $C_\bullet$ in $D(A)$. The proof is finished once the reader contemplates the distinguished triangle

$$B_\bullet \otimes_A^{\mathbf{L}} M \to B \otimes_A^{\mathbf{L}} M \to C_\bullet \otimes_A^{\mathbf{L}} M$$

Namely, the first term is represented by $B_\bullet \otimes_A M$ which is sitting in homological degrees $(t-1), \ldots, 0$ in view of the fact that the terms in the Koszul complex $B_\bullet$ are free (and hence flat) A-modules. Whence $\mathrm{Tor}_i^A(B, M) = H_i(C_\bullet \otimes_A^{\mathbf{L}} M)$ for $i > t - 1$ and this is annihilated by a_1^c. Since $a_1^c B = I^c B$ and since the tor module is a module over B we conclude. □